

Electro Magnetic Field Theory(UE24EE242B)

Assignment Test 1

IV SEM EEE/ECE

Duration 1 HOUR

Answer all questions Q1 and Q7 are mandatory and each carry 7 Marks Remaining questions carry 6 Marks. $2 * 7 + 6 * 6 = 14 + 36 = 50$ Marks.

1.

Point charges 1 mC and -2 mC are located at $(3, 2, -1)$ and $(-1, -1, 4)$, respectively. Calculate the electric force on a 10-nC charge located at $(0, 3, 1)$ and the electric field intensity at that point.

2.

Five identical $15\text{-}\mu\text{C}$ point charges are located at the center and corners of a square defined by $-1 < x, y < 1, z = 0$.

- (a) Find the force on the $10\text{-}\mu\text{C}$ point charge at $(0, 0, 2)$.
- (b) Calculate the electric field intensity at $(0, 0, 2)$.

OR

Obtain the expression for Electric field at any point due to a finite line charge and an infinite line charge along z axis.

3.

Point charges Q_1 and Q_2 are, respectively, located at $(4, 0, -3)$ and $(2, 0, 1)$. If $Q_2 = 4$ nC, find Q_1 such that

- (a) The $\mathbf{E}$ at $(5, 0, 6)$ has no z-component
- (b) The force on a test charge at $(5, 0, 6)$ has no x-component.

OR

Obtain the expression for electric field at any point due to a circular ring.

4.

The finite sheet $0 \leq x \leq 1, 0 \leq y \leq 1$ on the $z = 0$ plane has a charge density $\rho_s = xy(x^2 + y^2 + 25)^{3/2}$ nC/m². Find

- (a) The total charge on the sheet
- (b) The electric field at $(0, 0, 5)$
- (c) The force experienced by a -1 mC charge located at $(0, 0, 5)$

OR

Obtain the expression for electric field at any point due to a Volume charge distributed surface using Gauss Law application.

5.

Determine $\mathbf{D}$ at $(4, 0, 3)$ if there is a point charge -5π mC at $(4, 0, 0)$ and a line charge 3π mC/m along the y-axis.

OR

Obtain the expression for electric potential at any point due to a dipole.

6.

Two point charges $-4\ \mu\text{C}$ and $5\ \mu\text{C}$ are located at $(2, -1, 3)$ and $(0, 4, -2)$, respectively. Find the potential at $(1, 0, 1)$ assuming zero potential at infinity.

OR

Obtain the expression for energy stored and Energy density from the work done expression.

7.

Given the potential $V = \frac{10}{r^2} \sin \theta \cos \phi$,

(a) Find the electric flux density $\mathbf{D}$ at $(2, \pi/2, 0)$.

(b) Calculate the work done in moving a $10\text{-}\mu\text{C}$ charge from point $A(1, 30^\circ, 120^\circ)$ to $B(4, 90^\circ, 60^\circ)$.

8.

Two dipoles with dipole moments $-5\mathbf{a}_z$ nC/m and $9\mathbf{a}_z$ nC/m are located at points $(0, 0, -2)$ and $(0, 0, 3)$, respectively. Find the potential at the origin.

OR

Three point charges -1 nC, 4 nC, and 3 nC are located at $(0, 0, 0)$, $(0, 0, 1)$, and $(1, 0, 0)$, respectively. Find the energy in the system.